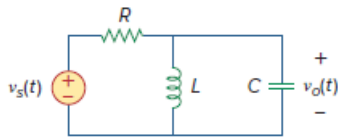


Problem

Refer to Fig. 10.69. If $v_s(t) = V_m \sin \omega t$ and $v_o(t) = A \sin(\omega t + \phi)$, derive the expressions for A and ϕ .

Figure 10.69:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.69 in the textbook.

Consider $v_o(t)$ is the voltage across the capacitor.

Draw the frequency domain equivalent circuit as shown in Figure 1.

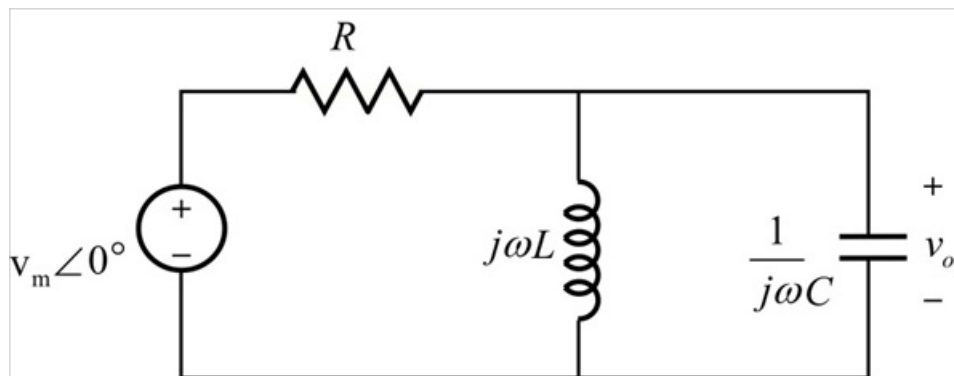


Figure 2

Step 2 of 4

Consider L and C are in parallel combination and determine the equivalent impedance.

$$\begin{aligned} Z_{eq} &= (j\omega L) \parallel \left(\frac{1}{j\omega C} \right) \\ &= \frac{(j\omega L) \left(\frac{1}{j\omega C} \right)}{j\omega L + \frac{1}{j\omega C}} \\ &= \frac{\frac{L}{C}}{j\omega L + \frac{1}{j\omega C}} \\ &= \frac{j\omega L}{1 - \omega^2 LC} \end{aligned}$$

Determine the voltage across capacitor v_o .

Apply voltage division at v_o .

$$v_o = \left[\frac{Z_{eq}}{R + Z_{eq}} \right] V_m \dots\dots (1)$$

Step 3 of 4

Replace $\frac{j\omega L}{1 - \omega^2 LC}$ for Z_{eq} in equation 1.

$$\begin{aligned} v_o &= \left[\frac{\frac{j\omega L}{1 - \omega^2 LC}}{R + \frac{j\omega L}{1 - \omega^2 LC}} \right] V_m \\ v_o &= \left[\frac{j\omega L}{R(1 - \omega^2 LC) + j\omega L} \right] V_m \\ |v_o| &= \frac{j\omega L V_m \angle 90^\circ}{\left[\sqrt{R^2 (1 - \omega^2 LC)^2 + \omega^2 L^2} \right] \angle \tan^{-1} \left[\frac{\omega L}{R(1 - \omega^2 LC)} \right]} \dots\dots (2) \end{aligned}$$

Step 4 of 4

Compare equation 2 with $v_o(t) = A \sin(\omega t + \phi)$.

Equation 2 can be written as,

$$|v_o| = \frac{j\omega LV_m}{\left[\sqrt{R^2(1-\omega^2 LC)^2 + \omega^2 L^2} \right]} \angle \left[90^\circ - \tan^{-1} \left[\frac{\omega L}{R(1-\omega^2 LC)} \right] \right]$$

By comparing,

$$A = \frac{j\omega LV_m}{\left[\sqrt{R^2(1-\omega^2 LC)^2 + \omega^2 L^2} \right]}$$

$$\phi = \left[90^\circ - \tan^{-1} \left[\frac{\omega L}{R(1-\omega^2 LC)} \right] \right]$$

Therefore, the values of A is $A = \frac{j\omega LV_m}{\left[\sqrt{R^2(1-\omega^2 LC)^2 + \omega^2 L^2} \right]}$ and the value of ϕ is

$$\phi = \left[90^\circ - \tan^{-1} \left[\frac{\omega L}{R(1-\omega^2 LC)} \right] \right].$$